

WANG Ruisi

Researcher (L6), SenseTime International | Multimodal Models · Visual Reasoning

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EXPERIENCE & EDUCATION

Researcher (L6) | SenseTime International, Singapore

Aug 2024 – Present

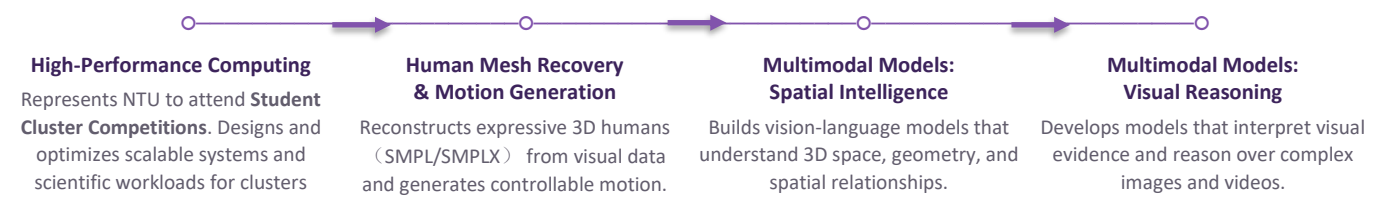
Promoted twice to Researcher (L6); earned annual performance ratings of A and A+.
Research multimodal foundation models, spatial intelligence, and visual reasoning.
Developed SenseNova-SI, achieving SOTA performance among open-source models across recent spatial-intelligence benchmarks.
Contributed to research accepted at ECCV 2026, ICML 2026, CVPR 2026, ICLR 2026, and TPAMI 2025.

Bachelor of Engineering (Computer Science) | Nanyang Technological University, Singapore

Aug 2020 – Jun 2024

Minor in Mathematics; NTU Science and Engineering Undergraduate Scholarship.
CGPA 4.83/5, Graduated with Honours (Highest Distinction).
Undergraduate Research Programme under Asst. Prof. Ziwei Liu.
Specializations: Artificial Intelligence, High-Performance Computing, and Data Science & Analytics.

RESEARCH INTEREST



SELECTED PUBLICATIONS

[1] Demystifying Video Reasoning

ECCV 2026

Ruisi Wang*, Zhongang Cai, Fanyi Pu, Junxiang Xu, Wanqi Yin, Maijunxian Wang, Ran Ji, Chenyang Gu, et al.

- Reveals that reasoning in diffusion-based video generation models primarily unfolds **across denoising steps**, rather than sequentially across video frames, introducing the **Chain-of-Steps (CoS)** mechanism.
- Identifies emergent reasoning behaviors, including **working memory**, **self-correction and enhancement**, and **perception-before-action**; shows that key reasoning computations are concentrated in the middle layers of the Diffusion Transformer.
- Introduces **Training-Free Ensemble (TFE)**, a proof-of-concept inference-time method that aggregates latent representations from multiple random seeds of the same model. TFE improves reasoning performance across LTX2.3, Wan2.1, and Wan2.2 on VBVR-Bench, V-Reason Bench, MME-CoF, and RISE-Video.

[2] A Very Big Video Reasoning Suite

ICML 2026

Maijunxian Wang*, Ruisi Wang*, Juyi Lin*, Ran Ji*, Thaddäus Wiedemer, Qingying Gao, Dezhi Luo, Yaoyao Qian, et al.

- Introduces the **Very Big Video Reasoning (VBVR) Dataset**, comprising over one million video clips across 200 curated reasoning tasks.
- Presents **VBVR-Bench**, a verifiable evaluation framework with **rule-based, human-aligned scorers**, enabling **reproducible and interpretable diagnosis**
- Enables large-scale study of **video reasoning scaling behavior**, revealing early evidence of **emergent generalization to unseen reasoning tasks**.

[3] SenseNova-SI: Scaling Spatial Intelligence with Multimodal Foundation Models

CVPR 2026

Zhongang Cai*, Ruisi Wang*, Chenyang Gu*, Fanyi Pu*, Junxiang Xu*, Yubo Wang*, Wanqi Yin*, Zhitao Yang*, et al.

- Introduces **SenseNova-SI-8M**, a systematically curated dataset of **8 million diverse samples** organized under a principled taxonomy spanning five core spatial capabilities: Metric Measurement, Spatial Relations, Mental Reconstruction, Perspective-taking, and Comprehensive Reasoning.
- Develops the **SenseNova-SI** family by scaling spatial data on established multimodal foundation models, achieving state-of-the-art performance among comparable open-source models across eight spatial intelligence benchmarks while preserving strong general multimodal capabilities.
- Provides systematic analysis of **spatial data scaling**, **emergent cross-task generalization**, **robustness to overfitting and language shortcuts**, and **spatial Chain-of-Thought (CoT)**, revealing that conventional text-based CoT does not reliably outperform simple QA-style data scaling.

[4] The Quest for Generalizable Motion Generation: Data, Model, and Evaluation

ICLR 2026

Jing Lin*, Ruisi Wang*, Junzhe Lu*, Ziqi Huang, Guorui Song, Ailing Zeng, Xian Liu, Chen Wei, et al.

Transfers video-generation priors to motion generation through ViMoGen-228K, the ViMoGen and ViMoGen-light model family, and the MBench benchmark. The framework expands motion diversity and improves generalization, prompt fidelity, and motion quality in automatic and human evaluation.

[5] SMPLeSt-X: Ultimate Scaling for Expressive Human Pose and Shape Estimation

TPAMI 2025

Wanqi Yin*, Zhongang Cai*, Ruisi Wang, Ailing Zeng, Chen Wei, Qingping Sun, Haiyi Mei, Yanjun Wang, et al.

Studies systematic data and model scaling for expressive human pose and shape estimation across 40 datasets and up to 10 million training instances. The resulting generalist foundation models transfer well to unseen environments and achieve state-of-the-art results across seven benchmarks.

* Equal contribution where indicated. Author lists are abbreviated.

COMPETITIONS & AWARDS

HPC AI Innovation Challenge 2023 — 1st Place	Jul 2024
Developed and evaluated a high-performance AI solution using national supercomputing resources. Presented the winning work at the SCA 2025 HPCIC Winners Showcase.	
Hack For Cities 2023 — 1st Place	Sep 2023
Designed and pitched a quantum-computing approach to urban mobility challenges, translating technical concepts into a city-scale solution.	
ISC2023 Student Cluster Competition — 1st Place	Jun 2023
Led the NTU team to first place against 19 teams worldwide. Built and maintained the team's cluster; applied optimization skills, techniques to complete real-world scientific workloads including FluTAS, POT3D, and Quantum ESPRESSO.	
SC2022 Student Cluster Competition — Overall Runner-up & Best Poster	Nov 2022
Achieved second place overall in an international student cluster competition and received the Best Poster award for communicating the team's system and optimization work.	
ISC2022 Student Cluster Competition — 6th Overall	Jun 2022
Placed sixth overall while operating and optimizing a student-built cluster for scientific workloads at the International Supercomputing Conference.	
Dean's List – Computer Science — Top 5% of cohort in academic performance	Aug 2022
NTU President Research Scholar — Accomplished with Merit in URECA Undergraduate Research Programme.	Jun 2022
35th Chinese Physics Olympiad — National 2nd Prize & Provincial 1st Prize	2018
Chinese Academy of Sciences Comprehensive Scientific Research — Excellent Award	2018

SELECTED RESEARCH & PROJECTS

SenseNova-U1 SenseTime International	Tech Report on arxiv 2026
- Contributed to the spatial intelligence capabilities of SenseNova-U1, surpassing strong baselines such as Qwen3.5 on key benchmarks including VSI-Bench. - SenseNova-U1 pushes unified multimodal modeling toward an encoder-free paradigm, eliminating both the pretrained visual encoder and VAE to directly unify pixel-level perception and generation within a single architecture.	
MMHuman3D / 3D Human Pose and Shape Estimation URECA, NTU	Jun 2022 – 2024
Contributed to the MMHuman3D open-source codebase under Asst. Prof. Ziwei Liu; implemented and maintained CLIFF (Carry Location Information in Full Frames).	

ACADEMIC SERVICE AND TEACHING

Reviewing: International Journal of Computer Vision (IJCV); NeurIPS, ICLR, ICML, and other academic conferences.

Talk: SCA 2025 – HPCIC Winners Showcase Track; presented the winning HPC AI solution.

Teaching: Peer Coach, NTU School of Computer Science and Engineering (now CCDS); supported students in moving beyond Academically At-Risk status.

LEADERSHIP

OneArena 2022 Vice-President, Technical Department	Jan 2022 – May 2022
- Hosted the competition and developed code and games with RoboMaster EP. - Mentored and trained participants and club members through workshops. - Assisted with and performed robot dancing during University Welcome Freshmen 2022 as a COE Student Ambassador.	
TEDxNTU	
- Main Organizer, Communications Department	Oct 2022
- Sub-Committee Event Manager	Oct 2021
Organized events and invited innovative speakers to bring creative and inspiring ideas to the NTU community.	
IET NTU On Campus Club Event Director (Main Committee)	Aug 2020 – May 2022
Organized seminars and workshops to share technical knowledge.	
SM2 Scholars Executive Committee Member	Aug 2019 – May 2020
The Purple Parade 2019 Organising Committee Member	Oct 2019